

Recommendation System

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Examples - In e-commerce

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Examples - Mobile Apps

App Empfehlungen



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★★★★★ FREE



IMDb Movies & TV
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Definition - Problem domain

- Recommendation systems (RS) help to match **users** with **items**

Definition - Problem domain

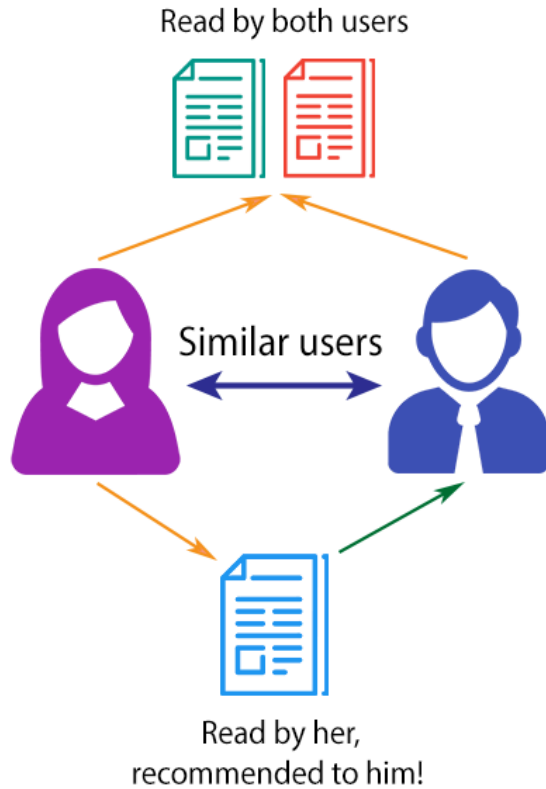
- RS are one of the **most successful and widespread applications** of machine learning technologies in business.

Two types of systems

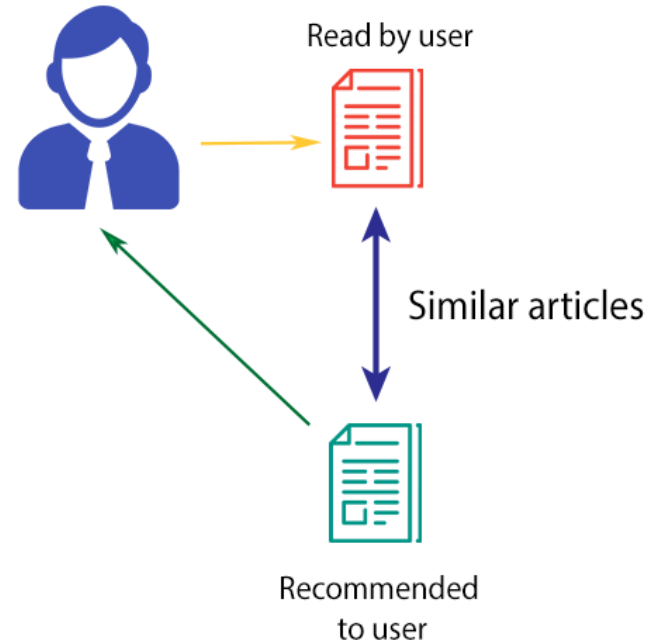
- **Content- Based Filtering:** Recommending to user A based on his/her existing profiles.
- **Collaborative Filtering:** Recommending to user A based on his/her community's profiles.

Two types of systems

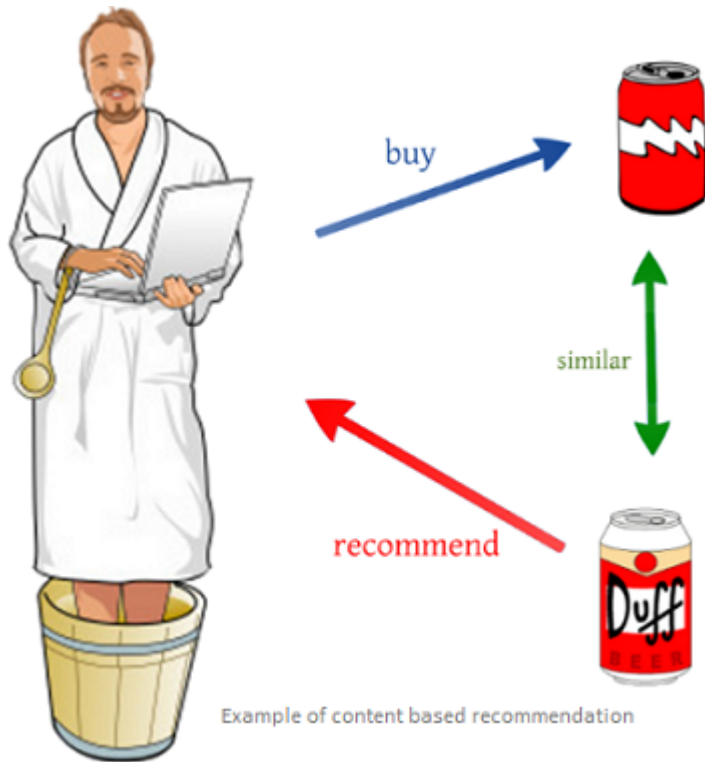
COLLABORATIVE FILTERING



CONTENT-BASED FILTERING



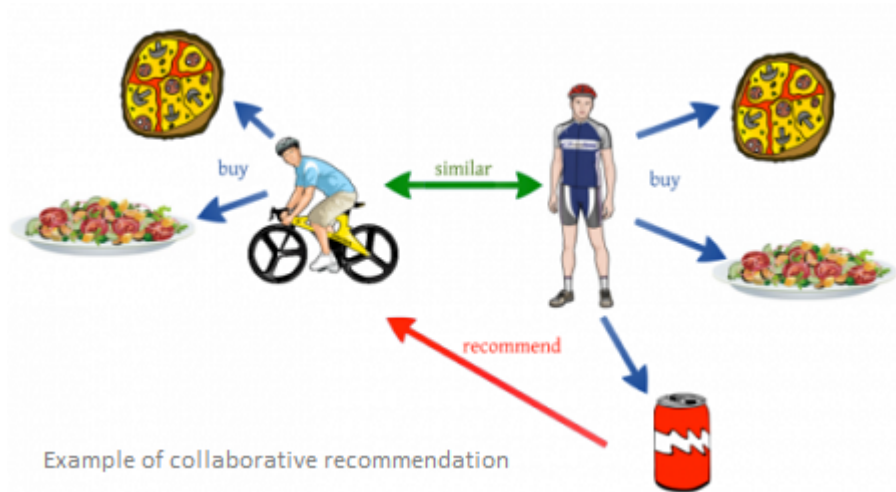
Content- Based Filtering



Content- Based Filtering

- Assume there are four categories of news A) Politics B) Sports C) Entertainment D) Technology
- User A who has read 10 articles related to Technology
- Recommend a new article in Technology for him to read.

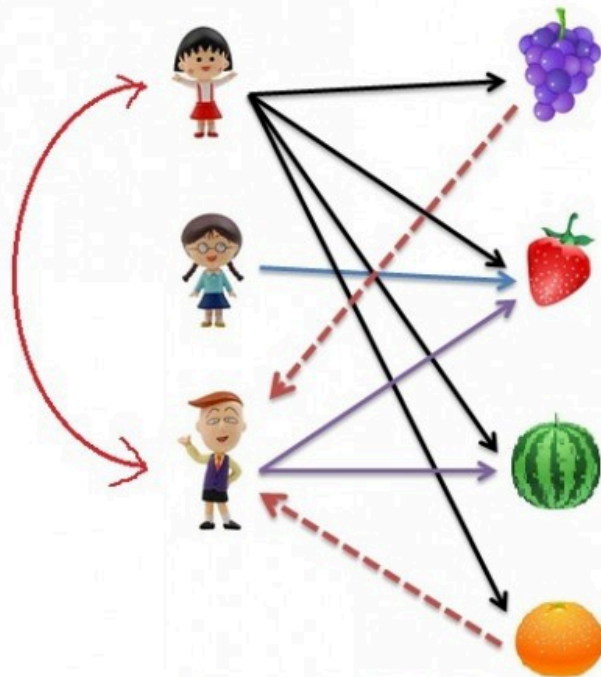
Collaborative Filtering



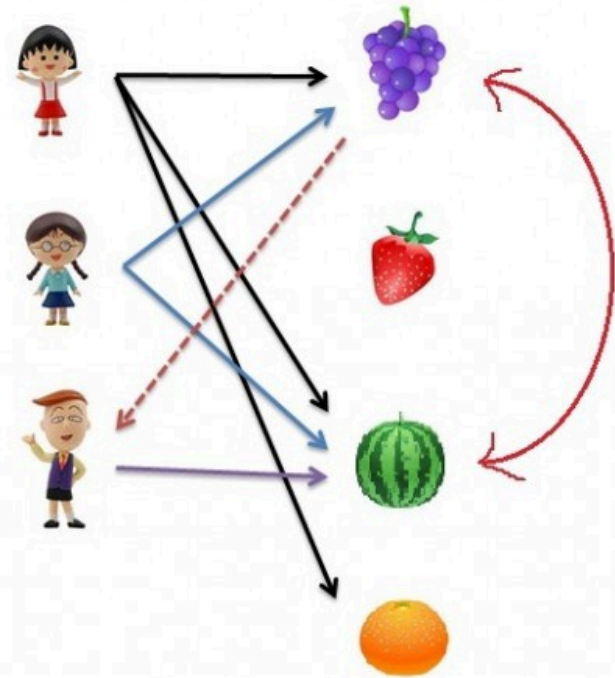
Collaborative Filtering

- Assume there are four categories of news A) Politics B) Sports C) Entertainment D) Technology
- User A who has read 10 articles related to Technology
- User B who has read **the same** 10 articles related to Technology and an X article in Sports.
- Recommend the article X to user A.

Collaborative Filtering: Two approaches



User-based filtering



Item-based filtering

Utility Matrix

- Utility Matrix contains ratings of users on items

	Item 1	Item 2	Item 3	Item 4	Item 5
Alice	5	3	4	4	???
User 1	3	1	2	3	3
User 2	4	3	4	3	5
User 3	3	3	1	4	4
User 4	1	5	5	2	1

- A **recommendation problem** turns into a **prediction problem**.
- Predict the rating of the new user on his/her new item.
- If the predicted rating of *Alice* on Item 5 are high (4 or 5), we will recommend Item 5 to her.

Nearest-neighbors (kNN)

- A "pure" CF approach and traditional baseline
- Using the utility as inputs
- Returns a ranked list of items based on rating predictions

Nearest-neighbors (kNN)

- **Assumptions**

- If users had similar tastes in the past they will have similar tastes in the future
- User preferences remain stable and consistent over time

User-based KNN

	Item 1	Item 2	Item 3	Item 4	Item 5
Alice	5	3	4	4	???
User 1	3	1	2	3	3
User 2	4	3	4	3	5
User 3	3	3	1	4	4
User 4	1	5	5	2	1

- Find k nearest neighbors of Alice.
- Use the average rating of the nearest neighbors on Item 5 as a prediction of Alice on Item 5.

User-based KNN

$$A1 = \sqrt{(5-3)^2 + (3-1)^2 + (4-2)^2 + (4-3)^2} = 3.60$$

A1 {

	Item 1	Item 2	Item 3	Item 4	Item 5
Alice	5	3	4	4	???
User 1	3	1	2	3	3
User 2	4	3	4	3	5
User 3	3	3	1	4	4
User 4	1	5	5	2	1

Let $A1$ is the distance from Alice to User 1 and so on. We have:

$$A1 = 3.60 \quad A2 = 1.41 \quad A3 = 3.60 \quad A4 = 5$$

- For 3NN, the predicted rating of Alice for item 5 is the average of ratings on item 5 of her 3 nearest neighbors, User 1, 2 and 3.
- Predicted rating of Alice on item 5 is: $(3+5+4)/3 = 4$.
- We will **recommend** item 5 to Alice.

Item-based KNN

	Item 1	Item 2	Item 3	Item 4	Item 5
Alice	5	3	4	4	???
User 1	3	1	2	3	3
User 2	4	3	4	3	5
User 3	3	3	1	4	4
User 4	1	5	5	2	1

- Find the k nearest neighbors of **Item 5**.
- The predicted rating of Alice on item 5 is the average rating of Alice on the nearest neighbors.

Item-based KNN

$$d_{54} = \sqrt{(1-2)^2 + (4-4)^2 + (5-3)^2 + (3-2)^2}$$

$$= \sqrt{5} = \approx 2.23$$

	Item 1	Item 2	Item 3	Item 4	Item 5
Alice	5	3	4	4	???
User 1	3	1	2	3	3
User 2	4	3	4	3	5
User 3	3	3	1	4	4
User 4	1	5	5	2	1

Let d_{54} be the distance of item 5 to item 4 and so on. We have

$$d_{54} = 2.23 \quad d_{53} = 5.19 \quad d_{52} = 5 \quad d_{51} = 1.41$$

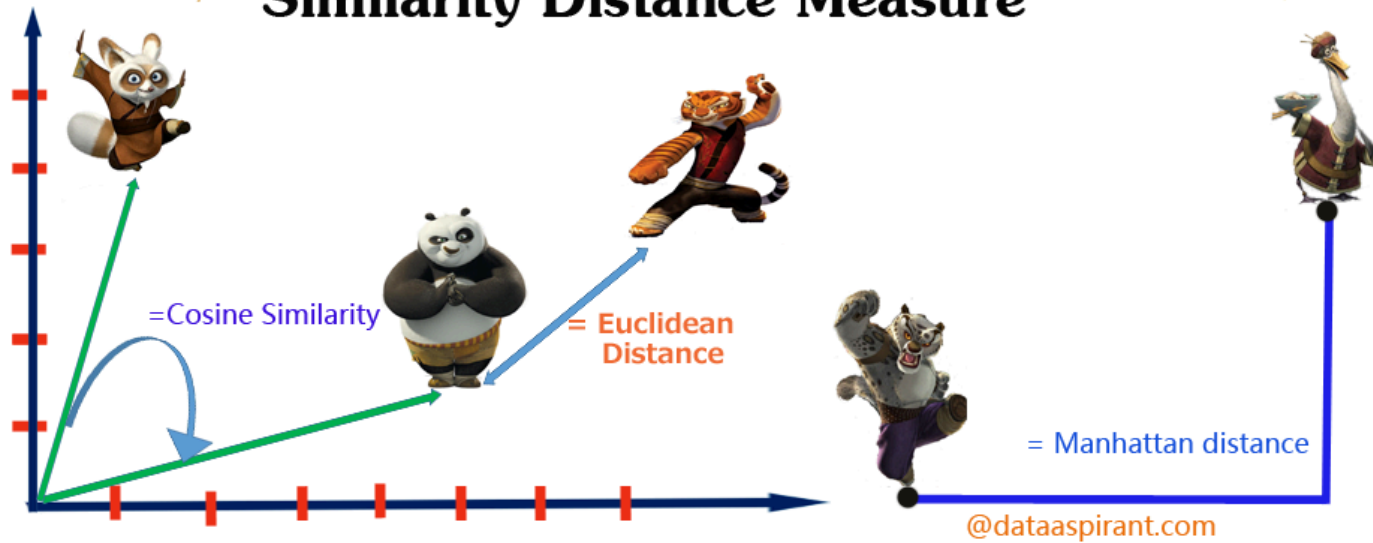
- For 3NN, the two nearest neighbors of Item 5 are Item 1,4 and Item 2.
- Predicted rating of Alice on Item 5 is the average of her ratings on Item 1, 4 and 2, which is $(5+4+3)/3 = 4$
- We **recommend** her to buy Item 5!

Similarity Measure

- Neighborhood can be decided by **similarity** measures
- Similarity can be measured as the inverse of the Distance
- The possible similarity values are between 0 and 1, where values near to 1 indicate a strong similarity.
- There are many distance measure
- There are many similarity measure

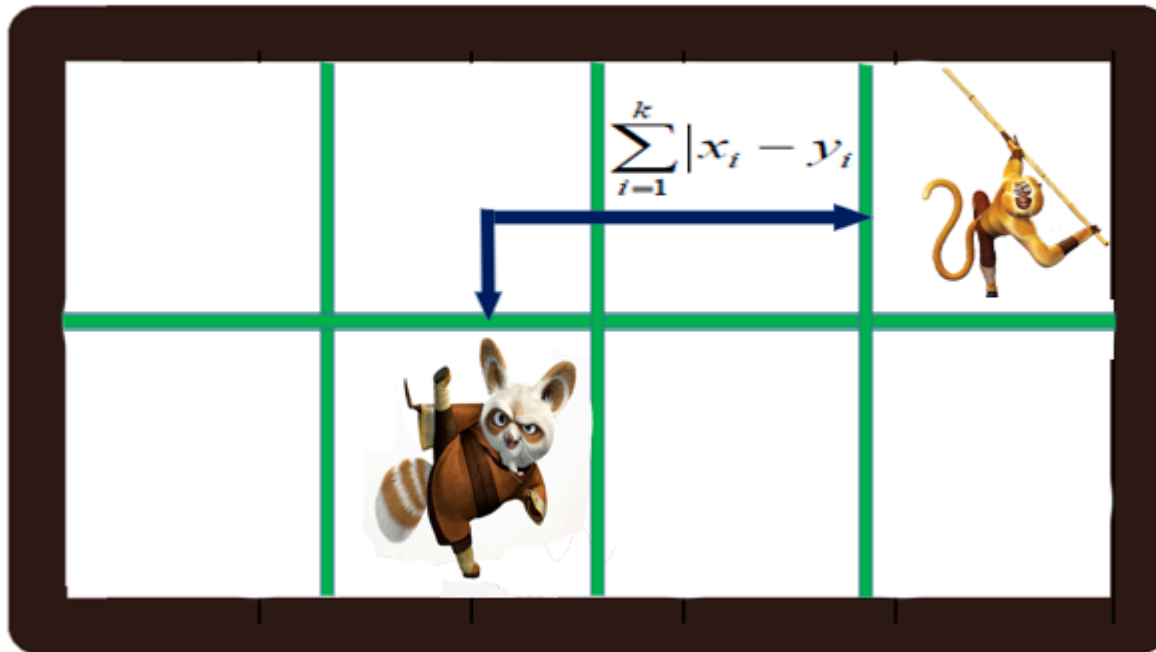
Similarity Measure

Similarity Distance Measure



Manhattan Distance

Manhattan Distance



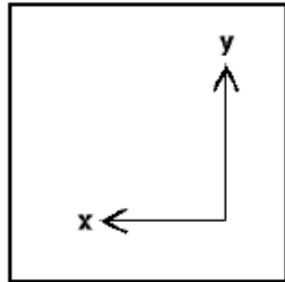
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- ManhattanDistance between Alice and User 1 (A1).

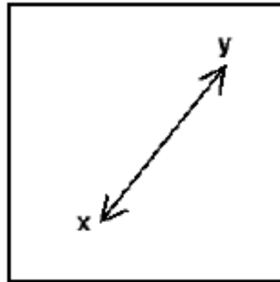
	Item 1	Item 2	Item 3	Item 4
Alice	5	3	4	4
User 1	3	1	2	3

$$A1 = |5 - 3| + |3 - 1| + |4 - 2| + |4 - 3| = 7$$

Manhattan vs. Euclidean



Manhattan

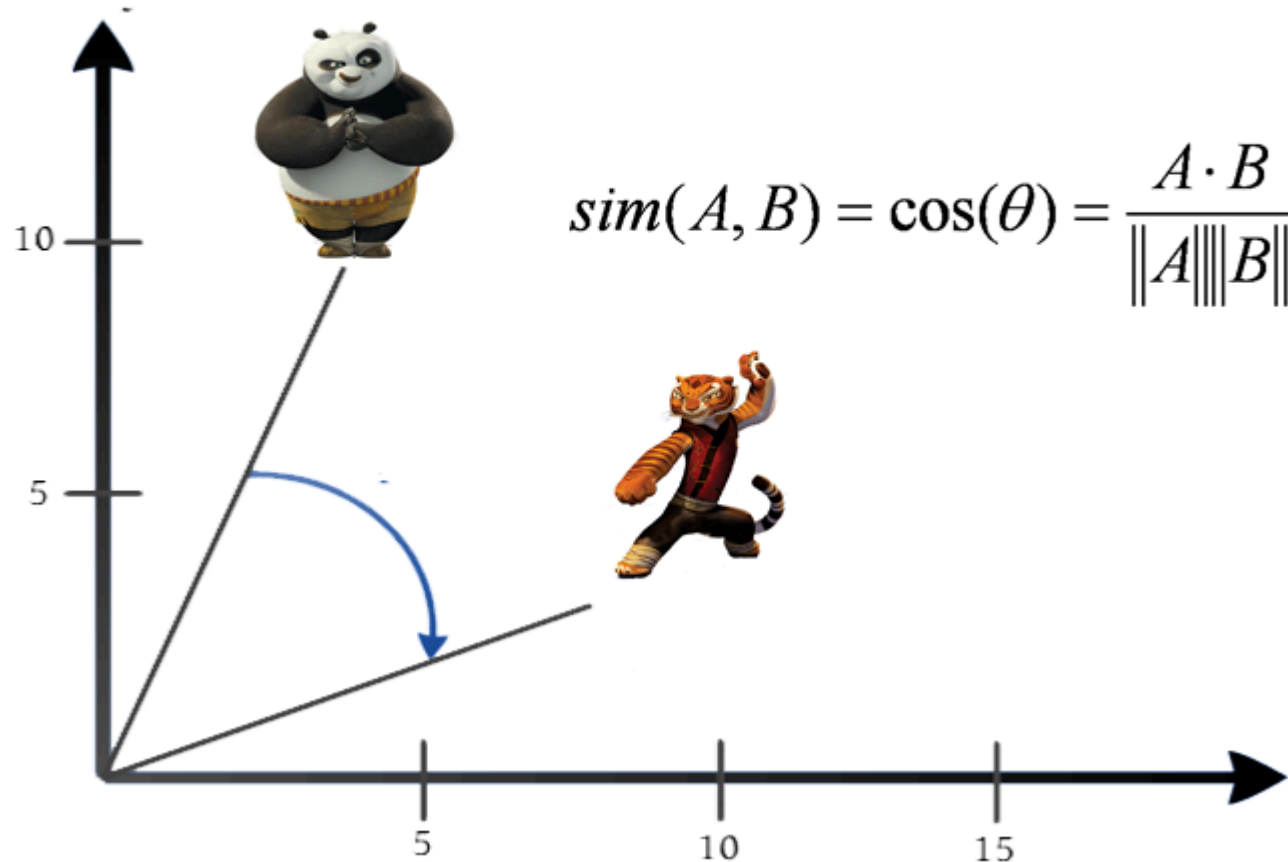


Euclidean

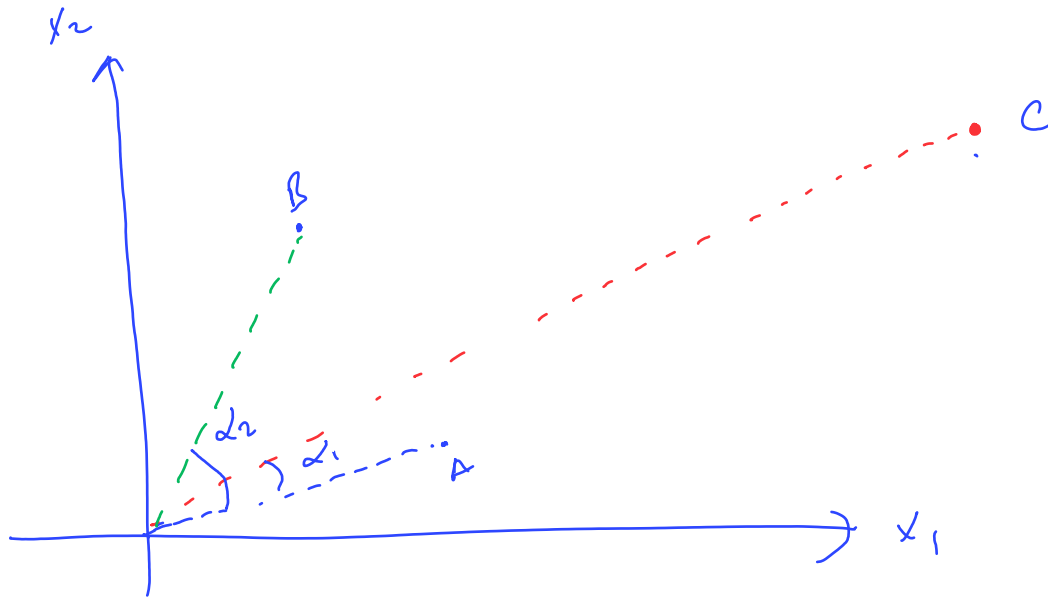


Cosine Similarity

Cosine Similarity



$$d(A, B) = \text{distance}$$



(*) E. distance: $d(A, C) > d(A, B) \Rightarrow B$ is more similar to A

(*) cos. sim: $d(A, C) < d(A, B) \Rightarrow C$ is more sim to A

Cosine Similarity Measure

- Cosine similarity between Alice and User 1 (S1).

	Item 1	Item 2	Item 3	Item 4
Alice	5	3	4	4
User 1	3	1	2	3

$$S1 = \frac{5 \cdot 3 + 3 \cdot 1 + 4 \cdot 2 + 4 \cdot 3}{\sqrt{5^2 + 3^2 + 4^2 + 4^2} \cdot \sqrt{3^2 + 1^2 + 2^2 + 3^2}} = 0.975$$

The Netflix Challenge

[Link](#)